SDET COURSE PROJECT 13 · PRIMER (READ BEFORE THE STUDY GUIDE)

From Scratch to Advanced: Every Idea Behind Our Project

Beginner-friendly build-up of all the mathematics: each step written out, each symbol explained, each result checked with a number from our own notebooks.

Why this document exists. The main study guide is complete but dense: it assumes you are comfortable with likelihoods, Fisher information, detection theory and feedback loops. This primer assumes only school algebra and basic calculus. It climbs in eight levels; every level ends with self-check questions (answers included) and the follow-ups a professor is likely to ask.

How every topic is written.

1. **Intuition** — the idea in plain words, before any symbol.
2. **Math, every step** — no step skipped; each line says *why* it follows from the previous one.
3. **Our number** — the same formula evaluated with values from our notebooks, so you can see it is real.
4. **Self-check** and **If the professor digs** — questions to test yourself, with answers.

Files: `docs/evaluation/primer_from_scratch.pdf` (this) → `study_guide.pdf` (full reference) → `question_bank.md` (all 46 questions) → `presentation_guide.md` (what to say per slide).

S. The study plan

S.1 The method (use it for every level)

1. **Read** the level once, slowly, with a pen.
2. **Close the PDF.** On blank paper, rewrite the boxed derivations from memory. Where you get stuck is exactly what you have not understood yet — reread only that step.
3. **Explain aloud** in plain words, as if to a first-year student (the "Feynman test"). If you need jargon to explain it, you do not own it yet.
4. **Answer the self-check** without looking; then check.

> **Why this works**
> In a viva the professor does not test whether you can *recognise* a derivation, but whether you can *produce* it and defend each step. Rewriting from a blank page is the only practice that matches that.

S.2 Full track (about 9–10 hours, can be split over two days)

Block	What	Time	Done when you can…
1	Level 0 — toolkit (dB, Gaussian, MSE, Q-function, Taylor, Cauchy–Schwarz)	60 min	prove MSE = bias² + variance and state Q(0), Q(1), Q(2)
2	Level 1 — signal model, E/N0, autocorrelation, bandwidth	60 min	explain every row of the "modelling choices" table and the 2 in E/N0 = $\Sigma s^2/(2\sigma^2)$
3	Level 2 — MLE and MMSE	60 min	derive J(θ) in three lines; derive the parabola formula; prove the posterior mean is MSE-optimal
—	break	15 min	
4	Level 3 — CRLB	75 min	prove the CRLB with Cauchy–Schwarz; compute it for our model; say three reasons it fails
5	Level 4 — the threshold effect, criteria, bootstrap	40 min	explain why a threshold is abrupt and why we need confidence intervals
6	Level 5 — ZZB (the hardest)	120 min	derive the ZZB formula and both limits on paper without notes
—	break	15 min	
7	Level 6 — DLL	75 min	derive $E - L \approx -2R'(d/2)\varepsilon$ and the loop variance $K\sigma^2/(2-K)$; explain N_eff = 16
8	Level 7 — reading our results	45 min	quote and interpret the headline numbers and each figure
9	Level 8 — proposal maths	45 min	derive the posterior-CRB recursion and its steady state
10	Question bank + slides aloud	60 min	give a 15-minute talk and answer 10 random questions

S.3 Short track (about 4 hours)

Must do	Why
Level 0.3–0.5, 0.7 (45 min)	MSE, Q-function, Cauchy–Schwarz are used in every later proof
Level 2.2, 3.2–3.4 (60 min)	MLE derivation + CRLB proof are the most-asked derivations
Level 5 complete (75 min)	the ZZB is the heart of the project
Level 6.2–6.4 (30 min)	S-curve and loop variance
Level 7 + "five numbers" (30 min)	you must quote results correctly

S.4 The night before and the last 30 minutes

- **Night before:** rewrite on one page: $J(\theta)$, CRLB = $1/(2(E/N_0)\beta^2)$, the ZZB integral, P_{min}, the two limits, the loop recursion. Sleep — recall is much worse when tired.
- **Last 30 minutes:** read only the "five numbers" (Level 7.1) and the "If the professor digs" boxes. Do not start anything new.

S.5 Where each level lives in the rest of the material

Primer level	Study guide section	Notebook / code
0 Toolkit	2	—
1 Model	3	00, `sdet/model.py`
2 MLE, MMSE	4	01, `sdet/estimators.py` (BatchMLE)
3 CRLB	5	03, `sdet/bounds.py` (crlb)
4 Threshold	6, 9	05 §E, 06
5 ZZB	7	04, `sdet/bounds.py` (zzb)
6 DLL	8	02, `sdet/estimators.py` (EarlyLateDLL)
7 Results	10	05, 06, 07
8 Proposal	14	not started (proposal only)

LEVEL 0

0. The mathematical toolkit

Everything later is built from these eight tools. None is difficult on its own; the later "hard" steps are just several of these used one after another.

0.1 Decibels

Intuition. Ratios in this project span a factor of 10,000 (−10 dB to 30 dB). Decibels turn multiplication into addition so such ranges are easy to talk about.

$$x_{\text{dB}} = 10 \log_{10} x, \qquad x = 10^{x_{\text{dB}}/10}.$$

dB	0	3	6	10	20	−10
linear	1	≈ 2	≈ 4	10	100	0.1

Our number. "MSE > 2 × CRLB" is the same as "MSE/CRLB > 3 dB". The MLE threshold 9.52 dB means $E/N_0 = 10^{0.952} \approx 8.95$. A "2.6 dB gap" means a factor $10^{0.26} \approx 1.8$ in signal energy.

0.2 Random variables, mean and variance

A **random variable** X is a quantity whose value depends on chance (here: noise). Its **mean** (expected value) is the long-run average, its **variance** the average squared distance from the mean:

$$\mathbb{E}[X] = \int x\, p(x)\, dx, \qquad \operatorname{var}(X) = \mathbb{E}\big[(X - \mathbb{E}X)^2\big] = \mathbb{E}[X^2] - (\mathbb{E}X)^2.$$

Two rules used constantly: **linearity** $\mathbb{E}[aX + bY] = a\,\mathbb{E}X + b\,\mathbb{E}Y$, and for **independent** X, Y: $\operatorname{var}(X+Y) = \operatorname{var} X + \operatorname{var} Y$, $\operatorname{var}(aX) = a^2 \operatorname{var} X$.

> **Proof of var X = E[X²] − (E X)²**
> Write $m = \mathbb{E}X$. Expand the square: $(X-m)^2 = X^2 - 2mX + m^2$. Take the mean of each term (linearity): $\mathbb{E}[X^2] - 2m\,\mathbb{E}X + m^2 = \mathbb{E}[X^2] - 2m^2 + m^2 = \mathbb{E}[X^2] - m^2$. ∎

0.3 Estimators, error, bias, MSE

An **estimator** $\hat{\theta}(\mathbf{r})$ is any rule that turns the received data $\mathbf{r}$ into a guess of the unknown θ (here the delay τ). Its **error** is $e = \hat{\theta} - \theta$. Three measures:

$$\text{bias} = \mathbb{E}[e], \qquad \text{variance} = \operatorname{var}(e), \qquad \text{MSE} = \mathbb{E}[e^2].$$

> **Proof: MSE = bias² + variance**
> This is exactly 0.2 applied to the random variable e: $\operatorname{var}(e) = \mathbb{E}[e^2] - (\mathbb{E}e)^2$, so $\mathbb{E}[e^2] = \operatorname{var}(e) + (\mathbb{E}e)^2$. ∎

Why we plot MSE, not variance

Below the threshold our estimators are **biased** (they get pulled toward wrong peaks and window edges). Variance alone would hide that error. MSE counts both parts — it is what a user of the delay estimate actually suffers.

An estimator is **unbiased** if $\mathbb{E}[e] = 0$ for *every* true value θ. This matters for the CRLB (Level 3).

0.4 The Gaussian distribution

Thermal noise is modelled as Gaussian: $w \sim \mathcal{N}(0, \sigma^2)$ has density

$$p(w) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\Big(-\frac{w^2}{2\sigma^2}\Big).$$

Facts we use: (a) a sum of independent Gaussians is Gaussian, means add, variances add; (b) $\mathbb{E}[w] = 0$, $\mathbb{E}[w^2] = \sigma^2$; (c) for N independent samples the joint density is the product of the individual densities, so the exponents add:

$$p(w_1, \ldots, w_N) = (2\pi\sigma^2)^{-N/2} \exp\Big(-\frac{1}{2\sigma^2}\sum_n w_n^2\Big).$$

(d) **Rotation invariance:** white Gaussian noise looks the same in every direction. If $\mathbf{u}$ is a unit vector ($\sum u_n^2 = 1$), the projection $\sum_n w_n u_n$ is again $\mathcal{N}(0, \sigma^2)$ — because its variance is $\sum_n u_n^2\sigma^2 = \sigma^2$. This single fact gives us the detection error in Level 5.

0.5 The Q-function

$Q(x)$ is the probability that a **standard** Gaussian $Z \sim \mathcal{N}(0,1)$ exceeds x:

$$Q(x) = P(Z > x) = \int_x^\infty \frac{1}{\sqrt{2\pi}} e^{-u^2/2}\, du.$$

x	0	1	2	3
Q(x)	0.5	0.159	0.023	0.0013

Properties: Q decreases from ½ to 0; its derivative is minus the Gaussian density, $Q'(x) = -\phi(x)$ with $\phi(x) = e^{-x^2/2}/\sqrt{2\pi}$; and if $X \sim \mathcal{N}(0, \sigma^2)$ then $P(X > a) = Q(a/\sigma)$ (divide by σ to make it standard).

A small integral we need later

$\int_0^\infty u\,Q(u)\,du = \frac{1}{4}$. **Proof** by integration by parts with $f = Q(u)$, $g' = u$ (so $g = u^2/2$, $f' = -\phi$): $\int_0^\infty uQ\,du = \big[\frac{u^2}{2}Q(u)\big]_0^\infty + \int_0^\infty \frac{u^2}{2}\phi(u)\,du$. The bracket is 0 at both ends (at ∞ Q decays faster than u² grows). The last integral is half of $\int_0^\infty u^2\phi\,du = \frac{1}{2}\mathbb{E}[Z^2] = \frac{1}{2}$ (half the line, by symmetry). So the result is $\frac{1}{2}\cdot\frac{1}{2} = \frac{1}{4}$. ∎

0.6 Taylor expansion

Near a point a, a smooth function is approximately a polynomial:

$$f(a+x) \approx f(a) + f'(a)\,x + \tfrac{1}{2} f''(a)\,x^2.$$

We use it three times: the CRLB is about the **curvature** (second derivative) of the correlation peak; the ZZB's high-SNR limit uses $1 - \rho(h) \approx \frac{1}{2}\beta^2 h^2$; the DLL's discriminator is linear near zero error.

0.7 Inner products and the Cauchy–Schwarz inequality

Treat N samples as a vector. The **inner product** $\langle \mathbf{a}, \mathbf{b} \rangle = \sum_n a_n b_n$ measures how much two signals "agree"; $\|\mathbf{a}\|^2 = \langle \mathbf{a}, \mathbf{a} \rangle$ is the energy. A **correlation** is just an inner product.

$$\textbf{Cauchy–Schwarz:} \qquad \langle \mathbf{a}, \mathbf{b} \rangle^2 \le \|\mathbf{a}\|^2 \, \|\mathbf{b}\|^2, \quad \text{equality iff } \mathbf{a} \propto \mathbf{b}.$$

> **Proof (two lines)**
> For any number t, $\|\mathbf{a} - t\mathbf{b}\|^2 \ge 0$, i.e. $\|\mathbf{a}\|^2 - 2t\langle \mathbf{a}, \mathbf{b} \rangle + t^2 \|\mathbf{b}\|^2 \ge 0$. A quadratic in t that is never negative has discriminant ≤ 0: $4\langle \mathbf{a}, \mathbf{b} \rangle^2 - 4\|\mathbf{a}\|^2 \|\mathbf{b}\|^2 \le 0$. ∎

The same inequality holds for random variables with $\langle X, Y \rangle = \mathbb{E}[XY]$: $(\mathbb{E}[XY])^2 \le \mathbb{E}[X^2]\,\mathbb{E}[Y^2]$. This is the whole CRLB proof (Level 3).

0.8 Swapping sum/integral and expectation

The mean of a sum (or integral) is the sum (integral) of the means. For non-negative quantities you may always swap the order (Tonelli/Fubini). We use this once, in the ZZB's "tail identity" (Level 5.1).

> **Self-check (answers below)**
> 1. What is 13 dB in linear terms? 2. An estimator always outputs 0. Bias? Variance? MSE (true θ = 1)? 3. If $X \sim \mathcal{N}(0, 4)$, what is $P(X > 2)$? 4. When does Cauchy–Schwarz hold with equality?
>
> **Answers.** 1. 10 × 2 = 20. 2. bias −1, variance 0, MSE 1. 3. σ = 2, so Q(2/2) = Q(1) ≈ 0.159. 4. When the two vectors are proportional — this is why the matched filter (replica ∝ derivative of the signal) is optimal.

> **If the professor digs**
> **"Can an estimator have smaller MSE than the CRLB?"** Yes, if it is biased: the constant estimator in question 2 has zero variance. The CRLB only restricts unbiased estimators — which is why it says nothing useful below threshold.

LEVEL 1

1. The problem and the signal model

1.1 The problem in words

A transmitter sends a known waveform s(t). It arrives late by an unknown delay τ and with noise added. We want to estimate τ. In GNSS, τ × speed of light = distance to the satellite, so a delay error of 0.1 chip on GPS C/A (1 chip ≈ 293 m) is ≈ 29 m of range error. **Everything in the project asks: how small can the delay error be, and when does it suddenly become large?**

1.2 Chips, BPSK and the pilot

The waveform is built from 64 **chips** $c_k \in \{+1, -1\}$ (BPSK = binary phase-shift keying: the sign of a pulse carries ±1). The chip sequence is random but **fixed and known** to the receiver (a "pilot"; ours is seed 2026). We measure time in chips, so the chip duration is 1.

$$s(t) = \sum_{k=0}^{63} c_k \, p(t - k).$$

1.3 The pulse p(t)

A rectangular chip has infinitely sharp edges, hence infinite bandwidth. Level 3 shows that the CRLB is $1/(2(E/N_0)\beta^2)$ with β the bandwidth — a rectangle would give CRLB = 0, a meaningless bound. So we use a **raised-cosine** pulse with roll-off α = 0.5:

- **Nyquist property:** p(0) = 1 and p(k) = 0 at every other integer → chips do not interfere at the sampling instants (checked: max |p(k≠0)| = $3 \cdot 10^{-17}$).
- **Band-limited:** its spectrum ends at (1+α)/2 = 0.75 cycles/chip.
- **Truncated** to |t| ≤ 6 chips, with a smooth taper over the last chip (a hard cut creates a tiny kink whose derivative spoils the CRLB/ZZB comparison; the taper fixed ZZB/CRLB → 0.997).

1.4 Sampling and the received signal

We take 4 samples per chip ($t_n = n/4$). Sampling theorem: we need more than 2 × 0.75 = 1.5 samples/chip; 4 gives a margin so derivatives (needed by the CRLB and DLL) are accurate.

$$r_n = s(t_n - \tau) + w_n, \qquad w_n \sim \mathcal{N}(0, \sigma^2) \text{ independent.}$$

The delay is applied **analytically** (we evaluate the formula of s at $t_n - \tau$) — no interpolation, so at high SNR there is no artificial error floor (noise-free MLE error was 0.02 % of the CRLB at 30 dB).

1.5 The SNR: E/N0, and where the "2" comes from

> **Derivation of E/N0 = Σs²/(2σ²)**
> In continuous time, a signal of energy E in white noise of two-sided spectral density N0/2 gives a matched-filter output SNR of **2E/No** (textbook result). In our discrete model the matched filter is the correlation $\sum_n r_n s_n$: its signal part is $\sum s_n^2$, its noise part $\sum w_n s_n$ has variance $\sigma^2 \sum s_n^2$ (rule 0.2), so its SNR is $(\sum s_n^2)^2/(\sigma^2 \sum s_n^2) = \sum s_n^2/\sigma^2$. Setting the two equal: $\sum s_n^2/\sigma^2 = 2E/N_0$, i.e.
>
> $$\frac{E}{N_0} = \frac{\sum_n s_n^2}{2\sigma^2} \qquad \Longleftrightarrow \qquad \sigma^2 = \frac{\sum_n s_n^2}{2\,E/N_0}.$$
>
> This is exactly `sigma2()` in `sdet/model.py` ; the notebook checks it (requested 10.0, measured 10.03).

Why total E/No and not per-chip? Both bounds depend on the *total* energy collected. The per-chip value is E/No ÷ 64 (≈ −18 dB lower).

1.6 The prior: what we know before looking

τ is drawn uniformly over a window of width $T_a = 4$ chips. A uniform variable on a width T_a has variance $T_a^2/12 = 16/12 = 1.33$ chip² (proof: $\int_{-T_a/2}^{T_a/2} x^2\,dx/T_a = T_a^2/12$). This is the error of "knowing nothing but the window" — the ceiling any sensible estimator should stay under. The ZZB needs this prior; the CRLB ignores it.

1.7 Autocorrelation: the shape behind everything

Intuition. Slide a copy of the signal against itself by h and measure how similar it is. If the shifted copy still looks very similar, noise can easily confuse the two delays.

$$\rho(h) = \frac{\sum_n s(t_n)\,s(t_n - h)}{\sum_n s(t_n)^2}, \qquad \rho(0) = 1.$$

Our pilot: the main lobe reaches zero at ±1.34 chips; beyond 1.5 chips the side-lobes are at most 0.13.

- The **curvature at the top** (how sharp the peak is) sets the small-error precision → CRLB.
- The **side-lobes and the window width** set how likely a far-away wrong delay is → threshold, ZZB.

1.8 Gabor (RMS) bandwidth β²

$$\beta^2 = \frac{\sum_n s'(t_n)^2}{\sum_n s(t_n)^2} \quad \Big(= \frac{\int (2\pi f)^2 |S(f)|^2 df}{\int |S(f)|^2 df} \Big).$$

It measures how fast the signal changes — "steepness per unit energy". **Ours:** 2.448 /chip² (time domain) vs 2.447 (frequency domain): the two formulas agree to 0.05 %. Over 400 random pilots β² ranges 1.77–4.27, which is why every method in a comparison must use the **same** pilot.

Link between ρ and β²: ρ″(0) = −β²

Differentiate ρ twice with respect to h (only the shifted copy depends on h): $\rho''(h) = \sum_n s(t_n)s''(t_n - h) / \sum s^2$. At h = 0: $\rho''(0) = \sum s\, s'' / \sum s^2$. Integration by parts (sum ≈ integral; s vanishes at both ends): $\int s\, s''\, dt = [s\, s'] - \int s'^2\, dt = -\int s'^2\, dt$. Hence $\rho''(0) = -\beta^2$. Also ρ′(0) = 0 (ρ is even, maximal at 0). By Taylor (0.6):

$$1 - \rho(h) \approx \tfrac{1}{2}\beta^2 h^2 \quad \text{for small } h.$$

We will need this in the ZZB's high-SNR limit.

Self-check

1. Why not rectangular chips? 2. Why 4 samples per chip, not 1? 3. What is the prior variance for Ta = 16? 4. Which feature of ρ(h) does the CRLB see, and which does it miss?

Answers. 1. Infinite bandwidth → β² = ∞ → CRLB = 0. 2. Need > 1.5 samples/chip for a 0.75-cycle/chip spectrum; 4 gives accurate derivatives. 3. 256/12 = 21.3 chip². 4. It sees the curvature at the peak; it misses the side-lobes and the window.

If the professor digs

"Your first Sprint-1 script used an m-sequence and a fixed delay. Why change?" The ZZB is a Bayesian bound: it needs τ to be random with a prior. With a fixed delay there is nothing to average over. One model for all methods means theory and simulation always describe the same experiment.

"Is a uniform prior realistic?" It is the natural model after coarse acquisition: we know τ only to within a search cell. Notebook 07 shows how the results move with its width.

LEVEL 2

2. The estimators: maximum likelihood and MMSE

2.1 The likelihood idea

Intuition. For each candidate delay θ, ask: "if the true delay were θ, how probable is the data I actually received?" Pick the θ that makes the data most probable. That number, as a function of θ, is the **likelihood** $p(\mathbf{r};\theta)$.

2.2 Deriving the MLE (three lines)

If the delay is θ, the noise must have been $w_n = r_n - s(t_n - \theta)$. Write $s_\theta[n] = s(t_n - \theta)$. By 0.4(c):

$$p(\mathbf{r};\theta) = (2\pi\sigma^2)^{-N/2} \exp\Big(-\frac{1}{2\sigma^2} \sum_n \big(r_n - s_\theta[n]\big)^2 \Big).$$

> **Step by step**
>
> **Step 1 — take the log** (log is increasing, so the maximiser does not change): $\ln p = -\frac{N}{2}\ln(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_n (r_n - s_\theta[n])^2$.
>
> **Step 2 — expand the square:** $\sum (r_n - s_\theta)^2 = \sum r_n^2 - 2\sum r_n s_\theta + \sum s_\theta^2$.
>
> **Step 3 — drop what does not depend on θ** (the first log term and $\sum r_n^2$), and multiply by σ² > 0: maximising ln p is the same as maximising
>
> $$J(\theta) = \sum_n r_n\, s(t_n - \theta) \;-\; \tfrac{1}{2}\sum_n s(t_n - \theta)^2, \qquad \hat\theta_{\mathrm{ML}} = \arg\max_\theta J(\theta).$$

Meaning. The first term is the **correlation** of the data with a replica shifted by θ — a matched filter. The second term is half the replica energy; it is almost constant in θ, but we keep it so the estimator is the exact MLE. **σ² dropped out:** the MLE needs no knowledge of the SNR.

In code: `J = r @ templates.T - half_energy` (one row of templates per candidate θ).

2.3 Implementation: a grid, then a parabola

We evaluate J on a grid with step 1/64 chip over the prior window, take the best grid point, then fit a parabola through it and its two neighbours to land between grid points.

Derivation of the parabolic refinement

Put the best point at x = 0 and its neighbours at x = ±1 (units of the grid step), with values J_-, J_0, J_+. Fit $y = ax^2 + bx + c$: at x = 0, $c = J_0$; at ±1, $a \pm b + c = J_\pm$. Subtracting: $b = (J_+ - J_-)/2$; adding: $a = (J_+ + J_- - 2J_0)/2$. The top of the parabola is where $y' = 2ax + b = 0$:

$$x^* = -\frac{b}{2a} = \frac{1}{2}\,\frac{J_- - J_+}{J_- - 2J_0 + J_+}.$$

Exactly the line `delta = 0.5 * (jm - jp) / den` in `_ml_from_objective`. We require den < 0 (a maximum) and clip x^* to ±½.

2.4 Why the MLE reaches the CRLB at high SNR (preview of Level 3)

At high SNR the maximum is near the true τ, J is smooth there, and the estimate solves $J'(\hat\theta) = 0$. Linearising (Level 3.6) gives an error $e = -\sum w_n s_n' / \sum s_n'^2$, whose variance is exactly the CRLB. Our check: MSE/CRLB = 0.997 averaged over 14–30 dB.

2.5 The MMSE (posterior-mean) estimator

Intuition. The MLE picks the single highest peak — it *commits*. If two peaks are almost equally high, a smarter estimator **hedges**: it averages candidate delays weighted by how believable each is.

Step by step

Bayes' rule: $p(\theta \mid \mathbf{r}) \propto p(\mathbf{r};\theta)\,p(\theta)$. With a uniform prior, $p(\theta)$ is constant on the window, so the posterior ∝ the likelihood there.

Use Step 2 of 2.2: $p(\mathbf{r};\theta) \propto \exp\!\big(-\tfrac{1}{2\sigma^2}[\sum r^2 - 2\sum r s_\theta + \sum s_\theta^2]\big) \propto \exp\big(J(\theta)/\sigma^2\big)$ (the $\sum r^2$ part is a constant factor).

Posterior mean on the grid: with weights $w_i = \exp(J(\theta_i)/\sigma^2)$,

$$\hat\theta_{\text{MMSE}} = \frac{\sum_i \theta_i\, w_i}{\sum_i w_i}.$$

(In code we subtract max J before exponentiating, to avoid overflow — it cancels in the ratio.) Note: unlike the MLE, the MMSE **needs σ²**, i.e. it needs to know the SNR.

Why the posterior mean minimises the MSE

For any guess c made after seeing r: $\mathbb{E}[(\theta - c)^2 \mid \mathbf{r}] = \mathrm{var}(\theta \mid \mathbf{r}) + (\mathbb{E}[\theta \mid \mathbf{r}] - c)^2$ (this is 0.2/0.3 applied to the posterior). The first term does not depend on c; the second is zero exactly when $c = \mathbb{E}[\theta \mid \mathbf{r}]$. Averaging over r keeps the inequality. ∎

Why we added it. The brief does not ask for it (we label it an extension). It answers a key question: is the gap between the MLE and the ZZB because the *bound* is loose, or because the *MLE* is sub-optimal? The MMSE is the best any estimator can do on average, so MMSE – ZZB is pure bound looseness (2.59 dB) and MLE – MMSE is pure estimator sub-optimality (0.85 dB).

2.6 Why the MLE can be *worse* than knowing nothing

On pure noise the MLE's MSE was 3.09 chip², more than the prior variance 1.33. Reason: with no signal, J is just a random noise curve; its maximum is often at a **window edge** (20 % of the time exactly on an edge), and edges are 2 chips from the centre (squared error up to 4). The MMSE, by contrast, averages over the window and tends to the centre (error ≈ the prior variance). This is a nice illustration that "most likely" ≠ "smallest error".

Self-check

1. Why does σ^2 not appear in the MLE? 2. With values $J_- = 9$, $J_0 = 10$, $J_+ = 9.5$, where is the peak? 3. Why does the MMSE need the SNR? 4. Which estimator is better below threshold and why?

Answers. 1. It multiplies all of ln p equally, so it does not move the maximiser. 2. $x^* = \tfrac{1}{2}\cdot(9 - 9.5)/(9 - 20 + 9.5) = \tfrac{1}{2}\cdot(-0.5)/(-1.5) = +0.167$ grid steps (toward the higher neighbour, as expected). 3. The weights are $\exp(J/\sigma^2)$: σ^2 sets how sharply the posterior concentrates. 4. MMSE: it hedges between competing peaks instead of jumping fully to a wrong one.

If the professor digs

"Is your MLE exact, given the grid?" The grid is 1/64 chip and refined by a parabola; noise-free error is 0.02 % of the CRLB at 30 dB, so grid error is negligible at every SNR we report.

"Then why not always use the MMSE?" It needs the SNR and the prior, and costs exponentials; the MLE is the standard practical receiver. We use the MMSE as a *yardstick*.

LEVEL 3

3. The Cramér–Rao lower bound (CRLB)

3.1 Intuition first

If the likelihood has a **sharp** peak, a small change in θ makes the data much less likely, so the data "pins down" θ precisely. If the peak is **flat**, many θ explain the data equally well. The CRLB turns "sharpness of the peak" into a number: the **Fisher information** I(θ). The bound says variance ≥ 1/I.

3.2 Two ingredients: the score and its mean

The **score** is the slope of the log-likelihood: $S(\mathbf{r};\theta) = \partial \ln p(\mathbf{r};\theta)/\partial\theta$.

> **Fact A: the score has mean zero**
> $\mathbb{E}[S] = \int \frac{\partial_\theta p}{p}\, p\, d\mathbf{r} = \int \partial_\theta p\, d\mathbf{r} = \partial_\theta \int p\, d\mathbf{r} = \partial_\theta 1 = 0$. (Line by line: derivative of a log is derivative over the function; the p's cancel; swap derivative and integral; a density integrates to 1.)

The **Fisher information** is the variance of the score: $I(\theta) = \mathbb{E}[S^2]$.

3.3 The proof of the CRLB

> **Fact B: for an unbiased estimator, E[(θ̂ − θ)·S] = 1**
> Unbiased means $\int \hat\theta(\mathbf{r})\, p(\mathbf{r};\theta)\, d\mathbf{r} = \theta$ for every θ. Differentiate both sides with respect to θ ($\hat\theta(\mathbf{r})$ does not depend on θ): $\int \hat\theta\, \partial_\theta p\, d\mathbf{r} = 1$, i.e. $\mathbb{E}[\hat\theta S] = 1$. Subtract $\theta\, \mathbb{E}[S] = 0$ (Fact A): $\mathbb{E}[(\hat\theta - \theta)S] = 1$.

> **The CRLB itself**
> Apply Cauchy–Schwarz (0.7) to the random variables $\hat\theta - \theta$ and S:
>
> $$1 = \big(\mathbb{E}[(\hat\theta-\theta)S]\big)^2 \le \mathbb{E}[(\hat\theta-\theta)^2]\,\mathbb{E}[S^2] = \mathrm{var}(\hat\theta)\cdot I(\theta) \quad\Longrightarrow\quad \mathrm{var}(\hat\theta) \ge \frac{1}{I(\theta)}.\ \blacksquare$$
>
> **Equality** needs $\hat\theta - \theta \propto S$ — possible in our problem only approximately, at high SNR.

3.4 Computing it for our model

Step by step

Score. From 2.2, $\ln p = \text{const} - \frac{1}{2\sigma^2}\sum (r_n - s_\theta[n])^2$. Differentiate (chain rule; $\partial_\theta s(t_n-\theta) = -s'(t_n-\theta)$): $S = \frac{1}{\sigma^2}\sum_n (r_n - s_\theta[n])\,\partial_\theta s_\theta[n] = -\frac{1}{\sigma^2}\sum_n w_n\, s'(t_n-\theta)$, using $r_n - s_\theta[n] = w_n$ at the true θ.

Fisher information. $I = \mathbb{E}[S^2] = \frac{1}{\sigma^4}\sum_n\sum_m \mathbb{E}[w_n w_m] s'_n s'_m$. Noise samples are independent, so $\mathbb{E}[w_n w_m] = \sigma^2$ if n = m and 0 otherwise: $I = \frac{1}{\sigma^4}\sigma^2\sum_n s'^2_n = \frac{\sum s'^2_n}{\sigma^2}$.

Substitute σ² from 1.5, $\sigma^2 = \sum s^2/(2E/N_0)$, and the definition of β² (1.8):

$$\text{CRLB} = \frac{\sigma^2}{\sum s'^2_n} = \frac{\sum s^2_n}{2(E/N_0)\sum s'^2_n} = \frac{1}{2\,(E/N_0)\,\beta^2} \quad [\text{chip}^2].$$

Our numbers (β² = 2.448): at 10 dB, CRLB = 1/(2·10·2.448) = 0.0204 chip² (RMS error 0.14 chip); at 20 dB, 0.00204 chip² (RMS 0.045 chip). Every 10 dB of SNR divides the CRLB by 10 — a straight line on a log plot.

Reading the formula: more energy (E/N0) or a steeper signal (β²) → smaller error. That is why bandwidth matters in ranging systems.

3.5 Why the CRLB fails at low SNR — three separate reasons

1. **It assumes unbiasedness.** Below threshold our estimators are biased (−0.47 chip at 2 dB for a delay near the window edge), so the bound does not apply to them.
2. **It is local.** It only uses the curvature at the true peak, never the side-lobes or a far-away wrong peak — exactly where gross errors come from.
3. **It ignores the prior.** Below ≈ −8 dB the CRLB exceeds Ta²/12 = 1.33 chip², i.e. it "allows" more error than simply guessing the window centre. (Solve 1/(2·x·2.448) = 1.33 → x = 0.153 = −8.1 dB.)

3.6 Bonus: why the MLE attains the CRLB, and why any other linear rule is worse

Linearising the MLE

At high SNR, $\hat\theta = \tau + e$ with e small and $J'(\hat\theta) = 0$. With $r = s_\tau + w$ and the energy term constant, $J'(\theta) = -\sum_n r_n s'(t_n-\theta)$. Taylor in e around τ: $J'(\tau) = -\sum s\,s' - \sum w\,s' = -\sum w\,s'$ (since $\sum s\,s' = \frac{1}{2}\sum (s^2)' = 0$), and $J''(\tau) \approx \sum s\,s'' = -\sum s'^2$ (integration by parts, as in 1.8). So $0 = J'(\tau) + eJ''(\tau)$ gives

$$e = -\frac{\sum_n w_n s'_n}{\sum_n s'^2_n}, \qquad \text{var}(e) = \frac{\sigma^2\sum s'^2}{(\sum s'^2)^2} = \frac{\sigma^2}{\sum s'^2} = \text{CRLB}.$$

General lesson (used again for the DLL)

Any estimator that, near the truth, forms $e = \sum w_n g_n / \sum g_n s'_n$ with some weighting g has $\mathrm{var}(e) = \sigma^2 \|g\|^2 / \langle g, s' \rangle^2 \geq \sigma^2 / \|s'\|^2$ by Cauchy–Schwarz, with equality only if g ∝ s′. The MLE uses g = s′ (optimal). The DLL's early-minus-late uses a slightly different g, which is why its locked error is 1.07 × CRLB.

Self-check

1. Prove E[S] = 0 aloud. 2. Where exactly is unbiasedness used in the proof? 3. CRLB at 0 dB for our pilot? 4. What happens to the CRLB if β² doubles?

Answers. 1. See 3.2. 2. In Fact B (differentiating ∫θ̂p = θ). 3. 1/(2·1·2.448) = 0.204 chip². 4. It halves (−3 dB).

If the professor digs

"Is the Fisher information also the expected curvature?" Yes: $I = -\mathbb{E}[\partial_\theta^2 \ln p]$ (differentiate Fact A once more). Both forms give $\sum s'^2/\sigma^2$ here.

"Does the CRLB depend on τ?" Only through edge effects of the finite window; we checked β² at five delays (identical to 5 digits).

LEVEL 4

4. The threshold effect, and how we measure it

4.1 Where gross errors come from — one calculation

Intuition. J(θ) = (signal correlation) + (noise). At the true delay the signal part is a tall peak. At a far-away delay the signal part is small, but the noise wiggles J up and down. Occasionally a noise wiggle somewhere else is taller than the true peak, and the MLE jumps there: a **gross error**.

> **Probability that one particular wrong delay τ + h beats the truth**
> With $r = s_\tau + w$: $J(\tau) - J(\tau+h) = \underbrace{\sum s^2\,(1-\rho(h))}_{\text{signal margin}} + \underbrace{\sum_n w_n\,(s_\tau[n] - s_{\tau+h}[n])}_{\text{noise}}$ (the energy terms cancel because shifting does not change energy). The noise term is Gaussian with variance $\sigma^2\|s_\tau - s_{\tau+h}\|^2 = \sigma^2 \cdot 2\sum s^2(1-\rho)$ (expand the square as in 5.3). The wrong delay wins when the noise is below minus the margin:
>
> $$P = Q\left(\frac{\sum s^2(1-\rho)}{\sigma\sqrt{2\sum s^2(1-\rho)}}\right) = Q\left(\sqrt{\frac{\sum s^2}{2\sigma^2}(1-\rho)}\right) = Q\left(\sqrt{\tfrac{E}{N_0}\,(1-\rho(h))}\right).$$
>
> Remember this expression — it reappears as $P_{\min}$ in the ZZB.

Why the failure is abrupt. Q falls off like $e^{-x^2/2}$, so the chance of a jump collapses quickly as SNR grows. But when a jump happens it costs an error of order 1–2 chips (squared: 1–4 chip²), while the CRLB is only ≈ 0.02 chip² at 10 dB. So even a 1 % jump probability with a 1.5-chip error adds 0.01 × 2.25 ≈ 0.022 chip² — about the CRLB itself, doubling the MSE. **That is why our threshold sits near 9.5 dB:** notebook 01 measured jump probabilities of 0.8 % at 10 dB, 2.1 % at 8 dB, 8.7 % at 6 dB.

4.2 The three regions of every MSE-vs-SNR curve

Region	Our SNR range	What happens	Right theory
Asymptotic	above ≈ 12 dB	only small errors around the true peak	CRLB (MLE/CRLB = 0.997)
Threshold	≈ 6–12 dB	rare jumps dominate the MSE	ZZB (qualitatively)
No information	below ≈ −5 dB	estimate ≈ random in the window	prior variance Ta²/12

4.3 Two threshold definitions (fixed before looking at results)

- **MSE criterion:** the highest SNR at which MSE/CRLB first exceeds 2 (3 dB). Scanning *down* from high SNR, we interpolate between the two grid points that bracket the crossing, on a log scale.
- **Gross-error criterion:** the highest SNR at which P(|e| > 0.5 chip) first exceeds 1 %.

Two criteria because they answer different questions: MSE weighs big errors heavily; the gross-error rate counts how often you fail regardless of how badly. The DLL-vs-MLE answer differs between them (Level 7).

4.4 Monte Carlo, and why we need confidence intervals

We estimate MSE by averaging e^2 over N independent trials. Its standard error is std(e^2)/√N. Near the threshold e^2 is **heavy-tailed** (mostly ≈ 0.02, occasionally ≈ 4), so a few unlucky trials move the MSE a lot, and the threshold can shift by a dB. Hence 20,000 trials per point and a 0.5-dB grid in notebook 06.

> **The bootstrap, step by step**
>
> 1. At every SNR, draw N trials **with replacement** from the N errors we have (a "resample").
> 2. Recompute the threshold from the resampled errors.
> 3. Repeat 200–300 times → 200–300 threshold values.
> 4. The 2.5 % and 97.5 % percentiles form the 95 % confidence interval.
>
> Idea: the resamples mimic "running the experiment again". For a **difference** (e.g. DLL – MLE) we take the difference in each resample; if the interval excludes 0, the difference is significant.

> **An honest caveat you should know**
>
> The bootstrap captures trial-to-trial noise, not the error of interpolating across a coarse SNR grid. Notebook 05's quick table (2-dB grid) therefore has intervals that are too narrow — e.g. DLL gross threshold 8.0 dB there vs 8.81 dB [8.54, 9.06] in notebook 06. **Always quote notebook 06.**

> **Self-check**
>
> 1. Why is the transition abrupt rather than gradual? 2. What does "CI excludes 0" mean? 3. Why 20,000 trials?
>
> **Answers.** 1. Jump probability falls like a Gaussian tail, while each jump costs ~100× the CRLB. 2. The difference is statistically significant at 95 %. 3. MSE near threshold is driven by rare large errors; you need many trials to see enough of them.

> **If the professor digs**
>
> **"3 dB is arbitrary."** Yes, any threshold definition is a convention; we fixed it beforehand, report a second criterion, and the conclusions (ZZB optimistic by ≈ 2.6 dB; DLL ≈ MLE on MSE) hold under both. "MSE doubles" is a standard, interpretable choice.

5. The Ziv–Zakai bound (ZZB), derived from scratch

5.0 The big idea in one paragraph

Estimating a delay well implies being able to **tell two nearby delays apart**. If your estimate is usually within h/2 of the truth, you could use it to decide "is the delay φ or φ + h?". But detection theory gives the **smallest possible** error rate of *any* such decision, $P_{\min}$. So no estimator can have small errors more often than that limit allows. Summing this argument over every separation h gives a lower bound on the MSE. Because it looks at *all* separations — including large ones — it sees gross errors, unlike the CRLB.

The derivation has five steps. Learn them in order; each is short.

5.1 Step 1 — MSE as an integral of tail probabilities

> **The tail identity**
>
> For a non-negative random variable X: $X^2 = \int_0^X 2x\,dx = \int_0^\infty 2x\,\mathbf{1}\{X \geq x\}\,dx$ (the indicator is 1 while x ≤ X and 0 after). Take the mean of both sides and swap mean and integral (0.8); the mean of an indicator is a probability:
>
> $$\mathbb{E}[X^2] = \int_0^\infty 2x\,P(X \geq x)\,dx.$$
>
> Apply it to X = |e| and substitute x = h/2 (so dx = dh/2):
>
> $$\mathrm{MSE} = \mathbb{E}[e^2] = \int_0^\infty \frac{h}{2}\,P\left(|e| \geq \frac{h}{2}\right) dh.$$

Meaning. The MSE is a weighted sum, over all error sizes, of "how often is the error at least that large". Big errors carry big weights (h). Now we only need a lower bound on $P(|e| \geq h/2)$.

5.2 Step 2 — the best possible detector between two known signals

Set-up. Two equally likely hypotheses: $H_0 : \mathbf{r} = \mathbf{s}_0 + \mathbf{w}$ and $H_1 : \mathbf{r} = \mathbf{s}_1 + \mathbf{w}$.

Derivation of P_min = Q(d / 2σ)

(a) Best rule. With equal priors, the error-minimising rule picks the more likely hypothesis. The likelihoods are $\propto \exp(-\|\mathbf{r}-\mathbf{s}_i\|^2/2\sigma^2)$, so: **pick the signal closer to r**.

(b) When does it err under H0? When r is closer to s1: $\|\mathbf{r}-\mathbf{s}_1\|^2 < \|\mathbf{r}-\mathbf{s}_0\|^2$. Put $\mathbf{r} = \mathbf{s}_0 + \mathbf{w}$ and $\mathbf{d} = \mathbf{s}_1 - \mathbf{s}_0$, $d = \|\mathbf{d}\|$: $\|\mathbf{w}-\mathbf{d}\|^2 < \|\mathbf{w}\|^2 \iff \|\mathbf{w}\|^2 - 2\langle \mathbf{w}, \mathbf{d}\rangle + d^2 < \|\mathbf{w}\|^2 \iff \langle \mathbf{w}, \mathbf{d}/d\rangle > d/2$.

(c) Probability. $\langle \mathbf{w}, \mathbf{d}/d\rangle$ is the noise projected on a unit vector, so it is $\mathcal{N}(0,\sigma^2)$ (0.4 d). Hence $P(\text{error} \mid H_0) = Q\big(\frac{d/2}{\sigma}\big)$. By symmetry the same under H1, so

$$P_{\min} = Q\left(\frac{\|\mathbf{s}_1 - \mathbf{s}_0\|}{2\sigma}\right).$$

Picture: two points at distance d; the decision line is halfway; you err when noise pushes you more than d/2 toward the wrong side.

5.3 Step 3 — our two signals are the pilot at delays φ and φ + h

Distance between shifted copies

$d^2 = \sum_n \big(s(t_n-\phi-h) - s(t_n-\phi)\big)^2 = \sum s^2_{\phi+h} + \sum s^2_{\phi} - 2\sum s_\phi s_{\phi+h} = 2\sum s^2 - 2\rho(h)\sum s^2 = 2\sum s^2\,(1-\rho(h))$.

Then $\frac{d}{2\sigma} = \sqrt{\frac{2\sum s^2(1-\rho)}{4\sigma^2}} = \sqrt{\frac{\sum s^2}{2\sigma^2}(1-\rho)} = \sqrt{\frac{E}{N_0}(1-\rho(h))}$ (using 1.5), so

$$P_{\min}(h) = Q\left(\sqrt{\tfrac{E}{N_0}\,\big(1-\rho(h)\big)}\right).$$

(It does not depend on φ — we checked: it varies by ≈ 2·10⁻⁵. Notice it is exactly the "wrong peak wins" probability of 4.1.)

Our numbers. For a separation beyond the main lobe (ρ ≈ 0): at 0 dB, $P_{\min} = Q(1) = 0.16$; at 6 dB (E/N0 ≈ 4), $Q(2) = 0.023$; at 20 dB, $Q(10) \approx 10^{-23}$. Large errors are possible at low SNR and impossible at high SNR.

5.4 Step 4 — any estimator can be turned into a detector

Lower bound on P(|e| ≥ h/2)

(a) The detector. Given φ and h, decide "τ = φ + h" if $\hat{\theta} \geq \phi + h/2$, otherwise "τ = φ". If truly τ = φ, it errs iff $e = \hat{\theta} - \phi \geq h/2$. If truly τ = φ + h, it errs iff $e < -h/2$. With equal priors its error rate is the average of the two, and no detector beats $P_{\min}$:

$$P(e \geq \tfrac{h}{2} \mid \tau = \phi) + P(e \leq -\tfrac{h}{2} \mid \tau = \phi + h) \;\geq\; 2P_{\min}(h). \qquad (*)$$

(b) Average over the prior. Split $|e| \geq h/2$ into $e \geq h/2$ or $e \leq -h/2$: $P(|e| \geq \frac{h}{2}) = \int p(\tau)P(e \geq \frac{h}{2}|\tau)d\tau + \int p(\tau)P(e \leq -\frac{h}{2}|\tau)d\tau$. In the second integral rename τ = φ + h. With the uniform prior, p = 1/Ta on the window [0, Ta].

(c) Keep only φ where both φ and φ + h are inside the window (φ from 0 to Ta − h). The dropped parts are ≥ 0, so this can only make the right side smaller:

$$P(|e| \geq \tfrac{h}{2}) \;\geq\; \frac{1}{T_a}\int_0^{T_a - h} \big[\,(*)\text{'s left side}\,\big]\,d\phi \;\geq\; \frac{T_a - h}{T_a}\,2P_{\min}(h) = 2\Big(1 - \frac{h}{T_a}\Big)P_{\min}(h).$$

5.5 Step 5 — put it together

The ZZB

Insert 5.4 into 5.1. For h > Ta the bound in 5.4 is 0 (no room for two delays), so the integral stops at Ta:

$$\text{MSE} \;\geq\; \int_0^{T_a} \frac{h}{2}\cdot 2\Big(1 - \frac{h}{T_a}\Big)P_{\min}(h)\,dh = \boxed{\int_0^{T_a} h\left(1 - \frac{h}{T_a}\right) Q\left(\sqrt{\tfrac{E}{N_0}(1 - \rho(h))}\right) dh}$$

This is the bound in `sdet/bounds.py → zzb()` (plus valley filling, next).

Read each factor: h — big errors weigh more; (1 − h/Ta) — the fraction of the window where a pair at distance h fits; $P_{\min}(h)$ — how confusable delays h apart are at this SNR.

5.6 Valley filling (Bell et al. 1997)

$P(|e| \geq h/2)$ can only **decrease** as h grows. So for any η ≥ 0, the bound at h + η is also a bound at h. Taking the best one, replace $f(h) = (1 - h/T_a)P_{\min}(h)$ by $\mathcal{V}\{f\}(h) = \max_{\eta \geq 0} f(h + \eta)$ — it fills in dips of f caused by side-lobes. The bound can only get tighter. For our pilot the side-lobes are small (≤ 0.13) and valley filling changes the ZZB by < 0.01 %. In code: a running maximum from the right, `np.maximum.accumulate(f[:, ::-1])[:, ::-1]`.

5.7 Limit 1 — no signal: ZZB → Ta²/12

Step by step

E/No → 0 gives $Q(0) = \frac{1}{2}$ for every h. Then $\text{ZZB} = \frac{1}{2}\int_0^{T_a}\big(h - \frac{h^2}{T_a}\big)dh = \frac{1}{2}\Big(\frac{T_a^2}{2} - \frac{T_a^2}{3}\Big) = \frac{1}{2}\cdot\frac{T_a^2}{6} = \frac{T_a^2}{12}$, the prior variance. (f is decreasing here, so valley filling changes nothing.)

Meaning: with no data the best you can do is guess the window centre; the bound stays finite, unlike the CRLB. **Numerically:** 1.3323 vs 1.3333 (the tiny gap is the finite numerical SNR used).

5.8 Limit 2 — strong signal: ZZB → CRLB

> **Step by step**
>
> **(a)** At high SNR, $P_{\min}(h)$ is negligible unless h is tiny. For tiny h use 1.8: $1-\rho(h)\approx\frac{1}{2}\beta^2h^2$, so $P_{\min}(h)\approx Q(ah)$ with $a=\beta\sqrt{(E/N_0)/2}$.
>
> **(b)** For tiny h, $1-h/T_a\approx 1$, and extending the upper limit from Ta to ∞ adds only negligible area: $\mathrm{ZZB}\approx\int_0^\infty h\,Q(ah)\,dh$.
>
> **(c)** Substitute u = ah (h = u/a, dh = du/a): $\int_0^\infty \frac{u}{a}Q(u)\frac{du}{a}=\frac{1}{a^2}\int_0^\infty uQ(u)\,du=\frac{1}{4a^2}$ (box in 0.5).
>
> **(d)** $\frac{1}{4a^2}=\frac{1}{4\beta^2(E/N_0)/2}=\frac{1}{2(E/N_0)\beta^2}=\mathrm{CRLB}.$

Numerically: ZZB/CRLB = 0.9971 at 40 dB (→ 1). Getting both limits right is our main correctness check.

5.9 What happens in between — the integrand picture

At 20 dB the integrand $h(1-h/T_a)P_{\min}(h)$ lives only at h ≪ 1 chip: a local, CRLB-like picture. At 0 dB $P_{\min}\approx 0.16$ even at h = 2–3 chips, so large separations contribute: that is the bound "seeing" gross errors. The ZZB leaves the CRLB (ZZB > 2 × CRLB) at **5.95 dB**.

5.10 Why the ZZB is optimistic by ≈ 2.6 dB

Step 5.4 compares the truth with **one** alternative delay at a time (a *binary* test). A real estimator must reject **all** wrong delays in the window at once (an *M-ary* test): the chance that *any* of many noise peaks beats the truth is larger than the chance for one particular one. The ZZB never "adds up" these competitors, so it underestimates gross errors and puts the threshold too low in SNR. We measured this looseness directly against the best estimator: MMSE threshold − ZZB threshold = 8.54 − 5.95 = **2.59 dB**.

> **Self-check (do these on paper)**
>
> 1. Prove $\mathbb{E}[X^2]=\int_0^\infty 2xP(X\ge x)dx$. 2. Derive $P_{\min}=Q(d/2\sigma)$. 3. Show $d^2=2\sum s^2(1-\rho)$. 4. Why is there a factor (1 − h/Ta)? 5. Derive both limits. 6. Why can the ZZB not be tight in the threshold region?
>
> **Answers.** 1. 5.1. 2. 5.2. 3. 5.3. 4. Both delays of the pair must lie inside the prior window; that happens for a fraction (Ta − h)/Ta of the window. 5. 5.7, 5.8. 6. 5.10 — binary vs M-ary.

If the professor digs

"Your ZZB is 3.6 dB below your MLE threshold — is it wrong?" No: it reproduces both limits analytically and numerically, an independent re-implementation agrees to 0.12 %, and the MLE is above it at every SNR. 2.6 dB of the gap is the bound's known looseness (5.10); 0.85 dB is the MLE's sub-optimality (it commits to one peak; the MMSE hedges).

"Why does the ZZB need a prior?" Step 5.4(b) averages over τ — without $p(\tau)$ there is nothing to average, and the low-SNR limit (the prior variance) would not exist.

"Is this the ZZB of Bell et al. 1997?" Yes — the scalar version with valley filling, specialised to a known pilot in white Gaussian noise and a uniform prior.

LEVEL 6

6. The delay-locked loop (DLL)

6.1 Why a loop at all

The batch MLE re-searches the whole window every time — expensive. A real receiver instead **tracks**: it keeps a current estimate θ, measures how wrong it is using just two correlators, and corrects a fraction of that error, block after block. That is feedback, like a thermostat.

6.2 Early and late correlators, and the S-curve

Let the tracking error be ε = θ − τ (positive = our estimate is too late). Correlate the data with an **early** replica (shifted d/2 earlier than θ) and a **late** one (d/2 later), with spacing d = 0.5 chip. With R the signal's (unnormalised) autocorrelation, ignoring noise:

$$E = R(\varepsilon - \tfrac{d}{2}), \qquad L = R(\varepsilon + \tfrac{d}{2}), \qquad D(\varepsilon) = E - L.$$

> **Why E − L is proportional to the error (Taylor, 0.6)**
> Expand both around ε = 0: $R(\pm\frac{d}{2} + \varepsilon) \approx R(\pm\frac{d}{2}) + R'(\pm\frac{d}{2})\,\varepsilon$. R is even, so $R(-\frac{d}{2}) = R(\frac{d}{2})$ and R′ is odd, so $R'(-\frac{d}{2}) = -R'(\frac{d}{2})$. Subtract:
>
> $$D(\varepsilon) \approx \left[R(-\tfrac{d}{2}) - R(\tfrac{d}{2})\right] + \left[R'(-\tfrac{d}{2}) - R'(\tfrac{d}{2})\right]\varepsilon = -2R'(\tfrac{d}{2})\,\varepsilon.$$
>
> On the main lobe R is decreasing, so $R'(d/2) < 0$ and $k = -2R'(d/2) > 0$: D has the same sign as ε, and D/k ≈ ε.

The **S-curve** is D(ε)/k plotted over ε. Ours: slope 1.000 at zero; peak 0.44 at ε = 0.70 chip; correct sign up to **±1.80 chips** — the **pull-in range**. Inside it the loop is pushed back toward ε = 0; outside it, it is not.

Why d = 0.5? A trade-off: a narrow spacing approximates the derivative s′ better (less noise, closer to the CRLB) but shrinks the linear region; a wide one tolerates bigger errors but is noisier. 0.5 chip is a standard compromise.

6.3 The loop update and its noise — every step

Each update uses one block of 4 chips: $\theta_{j+1} = \theta_j - K\,(D_j - b_j)/\bar{k}$. In the linear region the corrected discriminator measures the error plus noise: $(D_j - b_j)/\bar{k} \approx \varepsilon_j + n_j$, with n_j zero-mean, variance σ_n^2, independent from block to block. Subtract τ from both sides:

$$\varepsilon_{j+1} = (1 - K)\,\varepsilon_j - K\,n_j.$$

> **(a) Without noise: the error shrinks geometrically**
> $\varepsilon_j = (1-K)^j \varepsilon_0$. With K = 2/17 ≈ 0.118, the error falls by a factor e every ≈ 1/K ≈ 8.5 updates. Noise-free check: final error ≈ 10^{-7} chip.

(b) With noise: the steady-state variance

In steady state $\mathrm{var}(\varepsilon_{j+1}) = \mathrm{var}(\varepsilon_j) = V$. Since n_j is independent of ε_j, variances add (0.2): $V = (1-K)^2 V + K^2\sigma_n^2$. Solve: $V\,[1-(1-K)^2] = K^2\sigma_n^2$, and $1-(1-K)^2 = 2K - K^2 = K(2-K)$, so

$$V = \frac{K\,\sigma_n^2}{2-K} = \frac{\sigma_n^2}{N_{\text{eff}}}, \qquad N_{\text{eff}} = \frac{2-K}{K}.$$

Meaning: the loop behaves like averaging N_{eff} independent blocks. With K = 2/17: $N_{\text{eff}} = (32/17)/(2/17) = 16$ blocks × 4 chips = **64 chips** — the same data length as the batch MLE. That is exactly why we chose K = 2/(N_eff + 1) = 2/17 (the familiar exponential-moving-average formula).

(c) Compare with the CRLB

One 4-chip block has 1/16 of the batch energy, and CRLB ∝ 1/energy, so an ideal per-block measurement has $\sigma_n^2 \approx 16 \cdot \mathrm{CRLB}_{64}$. Then $V \approx 16 \cdot \mathrm{CRLB}_{64}/16 = \mathrm{CRLB}_{64}$. Measured: 1.04–1.09 × CRLB. The exact linear theory gives 1.071: early-minus-late weights the data with a shape that is not exactly s′, and by Cauchy–Schwarz (3.6) any weighting other than s′ is worse. **This is by design of K — it is not a bound.**

Loop bandwidth. For this first-order loop $B_L = \frac{K}{2(2-K)T}$ with T = 4 chips per update: $B_L = \frac{1}{32\cdot 4} = 0.0078$/chip, and $1/(2B_L) = 64$ chips — the same "averaging time" stated differently. Each run is 160 updates = 10 N_eff = 640 chips, long enough to forget the starting error.

6.4 The engineering details (each was found by testing)

Detail	Problem without it	What we do
Acquisition first	prior is ±2 chips, pull-in only ±1.8 → some starts can never lock	coarse grid search (0.5-chip steps) on the first 64 chips, then track
Block replicas + offset b_j	with only 4 chips per block, E − L ≠ 0 even at zero error (depends on the chip pattern): MSE was 2 × CRLB at 30 dB	compute b_j from the **known** pilot (noise-free discriminator at zero error) and subtract it
Constant gain $\bar{k}$	per-block slopes vary 0.29–1.7× the average; dividing by a tiny slope amplifies noise	divide by the average slope
Windows follow θ	fixed windows bias the loop when θ drifts	shift the sample window with the current estimate

Is b_j cheating? No: it uses only the known pilot, never the true delay. With deliberately wrong timing in the calibration the results were identical.

6.5 Loss of lock — the DLL's own failure mode

If noise pushes ε beyond the pull-in range, the discriminator no longer points home: the loop drifts, and its MSE **grows with observation time** instead of settling. At 0 dB, 54 % of runs end with a gross error: 12 % because acquisition started outside the pull-in range, 42 % because the loop lost lock while tracking. A batch MLE has no such mechanism — each block is a fresh estimate.

6.6 Is the DLL-vs-MLE comparison fair?

Same pilot (repeated), same noise model, same SNR axis (batch E/N0 per 64 chips). But the DLL sees 640 chips — 10× the energy. So the 64-chip ZZB is **not** a bound for the DLL; we also computed the ZZB for the DLL's own 640 chips, and the DLL stays above it at every SNR, as it must.

> **Self-check**
>
> 1. Derive $D \approx -2R'(d/2)\varepsilon$. 2. Derive $V = K\sigma n^2/(2-K)$. 3. Why is N_eff = 16 the right choice? 4. What is pull-in range and why do we need acquisition?
>
> **Answers.** 1. 6.2 box. 2. 6.3(b). 3. It makes the loop average 64 chips, like the batch, so the locked comparison is fair. 4. The error range where the discriminator has the right sign (±1.80); the prior (±2) is wider, so a coarse search must bring us inside it first.

> **If the professor digs**
>
> **"Why first-order, not second-order?"** The delay is constant in our model; a first-order loop tracks a constant with zero steady-state error. A second-order loop is needed for a ramp (Doppler) — listed as a limitation.
>
> **"Is the DLL an estimator in the same sense as the MLE?"** Above threshold it is approximately a linear estimator (6.3), so the CRLB comparison is meaningful; below threshold it is a nonlinear feedback system, which is why loss of lock needs its own theory (the proposal).

LEVEL 7

7. Reading our results correctly

7.1 The five numbers to know by heart (pilot 2026, MSE criterion, notebook 06)

Quantity	Value	95 % CI
ZZB threshold (bound)	5.95 dB	— (deterministic)
MMSE threshold (best possible estimator)	8.54 dB	[8.35, 9.07]
MLE threshold	9.52 dB	[9.00, 9.66]
DLL threshold	9.38 dB	[8.72, 9.98]
MLE/CRLB above 14 dB	0.997	points 0.96–1.03

7.2 The gap, decomposed

From → to	Size	What it measures
ZZB → MMSE	2.59 dB	looseness of the bound (binary vs M-ary, 5.10)
MMSE → MLE	0.85 dB, CI [0.25, 1.24]	MLE sub-optimality: it commits to one peak (2.5)
ZZB → MLE	3.57 dB	total; across 5 random pilots 2.7–5.4 dB

Detail you may be asked: the point estimates differ by 9.52 − 8.54 = 0.98 dB; the 0.85 dB quoted is the mean of the bootstrap differences. Both lie inside the CI — the honest summary is "about 1 dB, significantly above 0".

7.3 DLL vs MLE — the answer depends on the question

Criterion	DLL − MLE	Reading
MSE > 2 × CRLB	−0.12 dB [−0.87, 0.59]	no significant difference
P(|e| > 0.5) > 1 %	−0.68 dB [−0.99, −0.40]	DLL fails less often near threshold (it sees 10× more data)
far below threshold	DLL MSE grows with time	loss of lock: the DLL is much worse

7.4 Pilot dependence and prior width

- **Five pilots:** MLE − ZZB ranges 2.7–5.4 dB (pilot 33 is the outlier). The ZZB threshold moves with each pilot's β^2 and side-lobes. **Measured, not explained** — say so if asked.
- **Prior width (notebook 07, 10,000 trials):** Ta = 1 → threshold **masked** (the window lies inside the main lobe, so there is no far-away peak to jump to); Ta = 4 → MLE 9.34 dB, ZZB 5.95 (gap 3.4 dB); Ta = 16 → MLE 13.01 dB, ZZB 11.56 (gap 1.5 dB). A wider window has more places for a noise peak, so the threshold rises by ≈ 3.7 dB. The shrinking gap is an observation we have not explained.

7.5 How to talk about each figure

Figure	One-sentence reading
MLE vs CRLB (nb 01/03)	On the CRLB above ≈ 12 dB; leaves it abruptly below ≈ 10 dB; capped by wrong-peak errors, not the CRLB, at low SNR.
ZZB vs CRLB (nb 04)	The ZZB equals the CRLB at high SNR, the prior variance at zero SNR, and bends at 5.95 dB.
Four curves (nb 05)	All methods agree above threshold; below it the MMSE tends to the prior variance, the MLE overshoots it, and the DLL drifts.
Threshold with CIs (nb 06)	The thresholds are located with error bars; the gap to the bound splits into 2.6 + 0.85 dB.
DLL failure split (nb 05)	DLL failures = acquisition outside pull-in + loss of lock; below threshold its MSE grows with time.
Prior width (nb 07)	Wider uncertainty → later threshold; 1-chip window masks it.

7.6 What is ours, and how to say it

Known (reproduced and verified): the MLE, CRLB, ZZB for delay, linear DLL theory; even the ZZB applied to GNSS code tracking (Emmanuele et al. 2012; Liu et al. 2016). **Our observations:** the gap decomposition against the MMSE, its pilot and prior-width dependence, the DLL failure split, the criterion-dependent DLL-vs-MLE answer. **Phrase:** "we found no prior work that…", never "we are the first".

8. The mathematics of the proposal

PROPOSED Nothing in this level has results yet. Learn it so you can explain the plan and defend its feasibility.

8.1 A moving delay

Real delays drift. Simplest model, a random walk: $\tau_k = \tau_{k-1} + u_k$, $u_k \sim \mathcal{N}(0, q)$. Now the question is how well we can **track** τ_k over time, and where trackers lose information.

8.2 The scalar Kalman filter in two steps

> **Predict, then update**
>
> **Predict:** before the new block, our uncertainty grows by the drift: $P^- = P + q$.
>
> **Update:** the new block gives a measurement of τ with noise variance R. Combining two independent Gaussian estimates, **information (1/variance) adds**: $\frac{1}{P^+} = \frac{1}{P^-} + \frac{1}{R}$, and the new estimate moves toward the measurement by the gain $K = \frac{P^-}{P^- + R}$: $\hat{x}^+ = \hat{x}^- + K(z - \hat{x}^-)$.

The DLL is this filter with a fixed gain: its update θ ← θ − K·D/k is $\hat{x}^+ = \hat{x}^- + K(z - \hat{x}^-)$ with "measurement" $z = \theta - D/k$. So a DLL whose K equals the steady-state Kalman gain is optimal in its linear regime (H1).

8.3 The posterior CRB (PCRB) recursion and its steady state

The Bayesian analogue of the CRLB for tracking has the same predict/update form in terms of information $J_k = 1/P_k$, with $J_D = 1/R$ the Fisher information of one block:

$$J_k = J_D + \frac{1}{q + 1/J_{k-1}}.$$

> **Steady state**
>
> Set $P_k = P_{k-1} = P$: $\frac{1}{P} = \frac{1}{R} + \frac{1}{q+P} \Rightarrow P = \frac{R(q+P)}{R+q+P} \Rightarrow P(R+q+P) = Rq + RP \Rightarrow P^2 + qP - qR = 0$. Take the positive root:
>
> $$P_\infty = \tfrac{1}{2}\Big(-q + \sqrt{q^2 + 4qR}\Big), \qquad K_\infty = \frac{P_\infty + q}{P_\infty + q + R}.$$

8.4 The optimal tracker on a grid

The state is one number, so the exact Bayes filter can be computed on a grid of delays: **predict** = convolve the current posterior with the Gaussian drift kernel N(0, q); **update** = multiply by the block likelihood ∝ exp(J(θ)/σ²) (Level 2.5); **normalise**. No particle-filter approximation is needed.

8.5 Mean time to lose lock (MTLL) as a Markov chain

Discretise the tracking error into bins. From the measured S-curve and the noise level, compute the probability of moving from each bin to each other bin in one update (matrix Q, restricted to the "locked" bins). Leaving the pull-in range is an absorbing "lost" state. The expected number of updates until absorption, t, solves $(I - Q)\,\mathbf{t} = \mathbf{1}$.

8.6 The four hypotheses, in one line each

- **H1** Above threshold, the DLL with K∞ and the grid Bayes tracker both reach the PCRB.
- **H2** The DLL leaves the bound earlier; conditioning on "still locked" removes most of the excess.
- **H3** The Markov chain built on our S-curve predicts the MTLL and the DLL's threshold.
- **H4** The posterior probability of |error| > 0.5 chip warns of loss of lock earlier than a classical lock detector (compared by ROC curves).

> **If the professor digs**
>
> **"What if H2 fails?"** Then the DLL's extra loss is not mainly loss of lock but its sub-optimal (fixed-gain, linear) processing even while locked — still a clear, publishable decomposition. Every hypothesis has a useful negative outcome.

F. One-page formula sheet (rewrite this from memory the night before)

Quantity	Formula
Model	$r_n = s(t_n - \tau) + w_n,\ w_n \sim \mathcal{N}(0, \sigma^2),\ \tau \sim \mathcal{U}(\text{width } T_a)$
SNR	$E/N_0 = \sum s_n^2/(2\sigma^2)$
Autocorrelation, bandwidth	$\rho(h) = \sum s(t_n)s(t_n - h)/\sum s^2;\quad \beta^2 = \sum s'^2/\sum s^2;\quad 1-\rho \approx \frac{1}{2}\beta^2 h^2$
MLE	$\hat{\theta} = \arg\max_\theta \ \sum r_n s(t_n - \theta) - \frac{1}{2}\sum s(t_n - \theta)^2$
Parabola	$x^* = \frac{1}{2}(J_- - J_+)/(J_- - 2J_0 + J_+)$
MMSE	$\hat{\theta} = \sum \theta_i w_i / \sum w_i,\ w_i = e^{J(\theta_i)/\sigma^2}$
CRLB	$\text{var} \geq 1/I,\ I = \sum s'^2/\sigma^2 \ \Rightarrow\ 1/(2(E/N_0)\beta^2)$
Tail identity	$\mathbb{E}[e^2] = \int_0^\infty \frac{h}{2} P(\lvert e\rvert \geq \frac{h}{2})\, dh$
Binary detection	$P_{\min} = Q(d/2\sigma) = Q(\sqrt{(E/N_0)(1-\rho(h))})$
ZZB	$\int_0^{T_a} h\, \mathcal{V}\{(1 - h/T_a)P_{\min}(h)\}\, dh$
Limits	$E/N_0 \to 0:\ T_a^2/12; \qquad E/N_0 \to \infty:\ \text{CRLB}\ \ (\int_0^\infty uQ(u)du = \frac{1}{4})$
DLL discriminator	$E - L \approx -2R'(d/2)\,\varepsilon$; pull-in ±1.80 chips
Loop	$\varepsilon_{j+1} = (1-K)\varepsilon_j - Kn_j;\ V = K\sigma_n^2/(2-K);\ N_{\text{eff}} = (2-K)/K = 16$
PCRB (proposal)	$J_k = J_D + 1/(q + 1/J_{k-1});\ P_\infty = \frac{1}{2}(-q + \sqrt{q^2 + 4qR})$